

Exploring Robotic Armrest for Supporting Mid-air Interactions in Room-Scale VR

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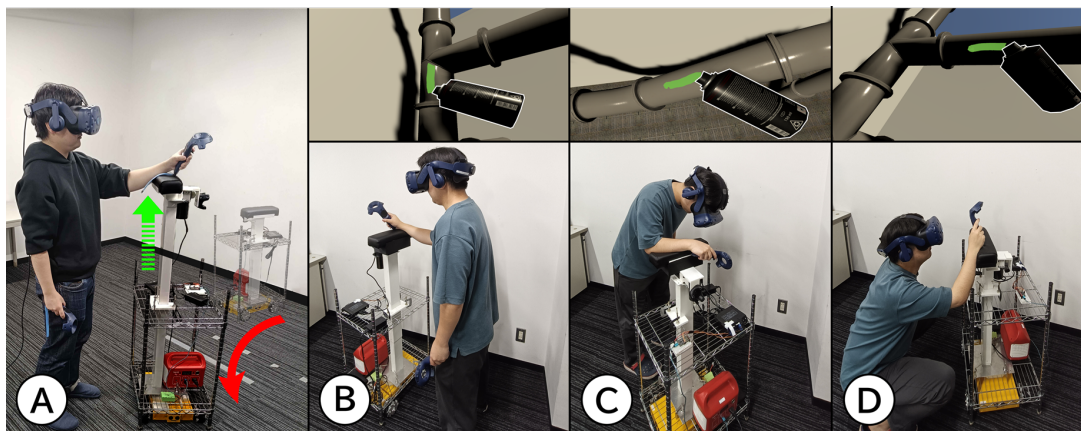


Figure 1: Users can interact stably with room-scale VR content through dynamic physical support provided by a robotic armrest. (A) The robotic armrest system dynamically adjusts the armrest's position, orientation, and height to accommodate users gestural interactions. Examples of robotic armrest interactions include: (B) eye-level input, (C) waist-level and downward-looking input, and (D) upward-looking input while in a lowered posture.

ABSTRACT

We explore a robotic armrest system that provides physical support for mid-air interactions to reduce arm fatigue in room-scale VR. The system dynamically adjusts the armrest's position, orientation, and height to center support around the user's arm while preserving input freedom and interaction consistently. We developed a prototype using actuators and a customized robotic station. A user study with a virtual handwriting task compared the robotic armrest to conventional mid-air interaction. Results show that the robotic armrest significantly reduces subjective physical workload while maintaining comparable input accuracy.

Index Terms: Omnidirectional robots, Gorilla Arm Syndrome, Arm fatigue, Ergonomics

1 INTRODUCTION

Room-scale VR allows users wearing an HMD (Head Mounted Display) to move freely within a tracked space and interact directly with virtual content using their bodies, providing a higher level of immersion than traditional VR systems. However, most current systems are based on mid-air interaction without physical support. Although this may be space-efficient, such interactions often cause rapid arm and shoulder fatigue during prolonged use and can reduce interaction accuracy [3].

Numerous interaction techniques have been proposed to mitigate fatigue in room-scale VR. Software-based approaches remap input performed at lower body levels to eye-level content, allowing more

comfortable postures [1, 5]. Although effective for indirect manipulation, their suitability for fine-grained tasks such as writing or drawing remains unclear. Wearable devices, such as exoskeleton like arm supports, can directly reduce arm load but often restrict movement and increase the burden of wearing and operating the device [3, 6]. Environmental robot-based supports, including encounter-type haptic displays [4] and mobile robotic props [2], provide localized haptic feedback but primarily enhance the realism at the point of interaction rather than support whole-arm movement.

Building on these insights, we explore a robotic armrest system that dynamically supports mid-air interaction in room-scale VR. The system provides an adjustable armrest that adapts its position, orientation, and height to the posture and the location of virtual content, reducing fatigue while maintaining freedom of movement.

This paper presents the design and implementation of a robotic armrest prototype. We also report the results of a user study that evaluates the robotic armrest's fundamental performance using a precise VR based writing task.

2 IMPLEMENTATION AND SPECIFICATIONS

Fig. 2 shows the implemented robotic armrest prototype. The system consists of an omnidirectional mobile robot, a linear actuator, a motion tracker, a battery unit, and an armrest.

The armrest is a lightweight, leather-based soft pad mounted on a rack-and-pinion linear actuator fixed to a metal frame, enabling motorized vertical adjustment over a 50 cm range.

Mobility is provided by a compact four-wheel-drive omniwheel platform mounted at the base. Both the mobile platform and the linear actuator are controlled by ESP32 micro controllers with wireless communication, allowing remote operation from a Windows PC. Power is supplied by a portable power station.

The room-scale VR environment is tracked using four SteamVR Base Station 2.0 units, covering an area of approximately 3.5 m

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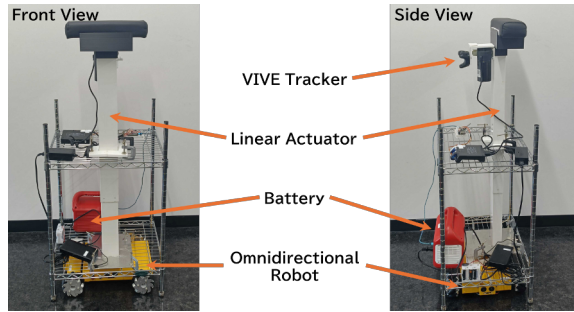


Figure 2: Robotic armrest prototype

$\times 3.5$ m. This setup accurately measures the position and orientation of the HMD and the controllers, as well as the position and height of the robotic armrest, which is equipped with an HTC VIVE Tracker 3.0. Based on the tracking data, the height of the robotic armrest's position is controlled using a linear actuator.

The prototype weighs approximately 20.5 kg. Because its movable components can reach the user's chest or shoulder level, we adhered to the collaborative robot safety standard ISO 10218-1:2025, which addresses close-proximity human-robot interaction. The mobile base speed was limited to 0.4 m/s, and the linear actuator speed to 0.023 m/s; both values are intentionally lower than a typical HMD user's walking speed (approximately 1 m/s). The robotic armrest supports a maximum payload of 70 kg, which is sufficient to support the weight of the users arms and, part of the upper body.

The system shows the armrest's real-time position and height, helping users avoid collisions and misorientation.

3 USER STUDY

We conducted a user study to investigate the effects of physical arm support provided by a robotic armrest on arm fatigue and input accuracy during mid-air interactions in a room-scale VR environment. Participants used VIVE handheld controllers to write characters on a virtual canvas. The performance of the robotic armrest was evaluated by comparing it with a conventional mid-air condition that provides no physical support. The experimental design was approved by our university's ethics review committee.

4 RESULT AND DISCUSSIONS

Twelve university and graduate students (3 females and 9 males) aged 18–23 years participated in the study; seven reported prior VR experience. Participants performed a standing, single-handed virtual handwriting task.

Figure 3 summarizes the task results. Asterisks (*) in the figure indicate statistically significant differences, as determined by an analysis of variance (ANOVA) applied to the collected data.

CR10 We found a significant main effect of the input method on CR10 scores. The perceived physical workload on the arm was significantly lower in the robotic armrest condition than in the mid-air condition ($F(1, 11) = 44.2, p < 0.01, \text{partial } \eta^2 = 0.80$). This result suggests that providing physical support during mid-air handwriting can substantially reduce arm fatigue, which is a well-known issue in prolonged VR interactions.

Input Accuracy Input accuracy was defined as the off-target time and calculated as the proportion of total writing time during which the pen trajectory deviated from the template (i.e., deviation time $\div$ total writing time $\times 100\%$). No statistically significant effect of input method on input accuracy was observed ($p > 0.05$). However, a slight numerical trend toward higher accuracy can be observed in Figure 3. This suggests that the physical support provided by the robotic armrest does not interfere with users' input

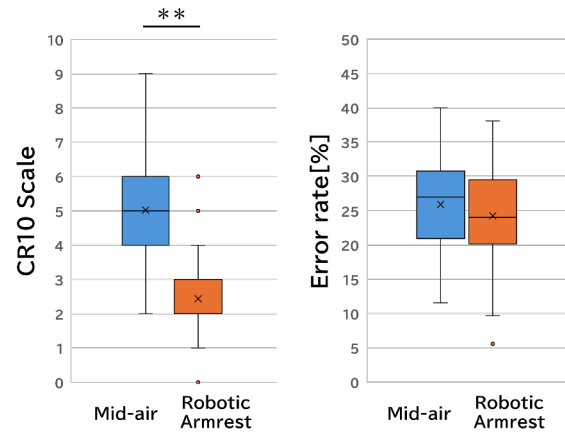


Figure 3: CR10 scores and error rate across input methods.

movements and does not degrade input accuracy during handwriting tasks.

Overall, the results demonstrate that the proposed robotic armrest effectively reduces perceived physical workload during virtual handwriting without negatively affecting input accuracy. However, the present study focused on a single-handed, single plane writing task, and several aspects remain unexplored.

As future work, we plan to explore a dual-arm configuration to support bi manual interaction, incorporate pressure sensors to measure and analyze load distribution, and evaluate the system in three-dimensional tasks that include vertical movements.

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